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FACULTY OF COMPUTER SCIENCE & MATHEMATICS
SOFTWARE ARCHITECTURE
CSE3433

Group No. 11

SERVICE-ORIENTED SMART CAMPUS SUPPORT SYSTEM

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Selected Architecture Style

Service-Oriented Architecture (SOA) is a client/server design approach in which an application consists of software services and software service consumers. Instead of building a single, tightly integrated system, SOA divides functionalities into smaller services, each responsible for a specific task such as data processing, user management, or payment handling. These services can be accessed and combined to support different applications, making the system more flexible and easier to maintain.

Problem Description

The old campus service management is very inconvenient as it is largely managed through manual processes and separate platforms for each service. They all rely on physical paperwork, registration forms, and payment methods. This traditional approach resulted in inefficiencies, long waiting time, and limited accessibility. Therefore, the overall management is not suitable anymore in this modern world.

There are several problems with the old campus service management which is the lack of a centralized system. This is because it uses multiple separate systems that are not connected to each other. For example, the student and staff must use different platforms for different services. This leads to data inconsistency, duplication of information, and difficulty in accessing accurate data.

Other than that, the administrative services are often slow due to manual processes as it relies on physical interactions. Tasks such as registration, fee payment, and handling inquiries require students to queue at service counters, resulting in long waiting times and delays. This reduces productivity and creates inconvenience for both students and administrative staff.

Lastly, facility management in many campuses is inefficient due to the absence of a proper digital system. Booking rooms, reserving labs, or reporting maintenance issues are often done manually or through informal methods. This leads to scheduling conflicts and delayed maintenance responses, ultimately affecting the overall campus experience.

System Objective

We propose on building a Smart Campus Support System with the implementation of Service-Oriented Architecture through APIs. This system aims to solve the problem stated and improve the users' experience that use the system.

In this system, all the system are centralized into a main pipeline. Multiple separate system is integrated into one big system while still being encapsulated inside their own modules. This enables the student and staff to use the same website but have different access level. This will also help in reducing data redundancy.

Other than that, the system has built-in registration and fee payment modules helping with administrative tasks such as registering courses, paying tuition fee or fines etc. This will increase the efficiency of the administrative task and increase the productivity of the student as they have more time on hand to practise.

Lastly, the system has its own booking module that has its own booking database. Any booking was done and saved by the system. This will reduce the chance of scheduling conflicts greatly.

Initial Functional Requirements

The system will include the following main functionalities:

1. User Management

- Students can register and log in
- Profile management

2. Facility Booking

- View available facilities (e.g., rooms, labs, halls)
- Book and cancel reservations

3. Complaint Management

- Submit complaints or issues
- Track complaint status

4. Notification System

- Receive announcements from administration
- Get alerts for booking status or complaint updates

5. Service Integration

- All services are connected through SOA
- Each module works independently but communicates with others

Team Member Roles

Sarah Amira Heri	Project Manager & System Analyst	• Plan and manage the overall project timeline • Gather and analyze system requirements • Define system architecture (SOA design) • Ensure the project meets objectives
Siti Aisyah Arwiah Binti Yundi	Backend Developer & Database Designer	• Develop core system functionalities (services in SOA) • Handle server-side logic and integration between services • Design and manage the database structure • Ensure data consistency and security
Elsa Fadlin Binti Mersing	Frontend Developer & Tester	• Design and develop the user interface • Ensure system is user-friendly and responsive • Conduct system testing and debugging • Report and fix errors to improve system quality